Surfacebound filaments

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1 Background and notation

1.1 Differential Forms

1.2 Euclidean Transformations

1.3

2 Kinematics of the Kirchhoff rod

The state of a Kirchhoff rod can be identified with with a map $\rho : S \to SE(3)$ from a 2-dimensional domain $S \subset \mathbb{R}^2$, parameterized by material coordinate s and time t, into the Special Euclidean group

$$SE(3) = \left\{ \begin{bmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{bmatrix} \in GL(4) \middle| Q \in SO(3), \boldsymbol{r} \in \mathbb{R}^3 \right\}. \tag{1}$$

The matrix $\rho(s,t) = \begin{pmatrix} Q & \mathbf{r} \\ 0 & 1 \end{pmatrix}$ represents the Euclidean transformation which maps an inertial frame $\boldsymbol{e}_i$ located at the origin onto the body's orthonormal material frame $\boldsymbol{q}_i(s,t) = Q(s,t)\boldsymbol{e}_i$ located at $\mathbf{r}(s,t)$.

2.1 Structure equations

Deformation and velocity fields (ψ_s, ψ_t) are components of an $\mathfrak{se}(3)$-valued one-form,

$$\psi = \psi_s \, \mathrm{d}s + \psi_t \, \mathrm{d}t = \rho^{-1} \, \mathrm{d}\rho \,, \tag{2}$$

where $\mathrm{d}\rho = \partial_s \rho \, \mathrm{d}s + \partial_t \rho \, \mathrm{d}t$ is the exterior derivative of ρ, and $(\mathrm{d}s \,, \mathrm{d}t)$ are 1-forms dual to the coordinate basis (∂_s, ∂_t). Repeated application of the exterior derivative gives

$$0 = \mathrm{d}^2\rho = \rho \, \mathrm{d}\psi + \mathrm{d}\rho \wedge \psi, \tag{3}$$

where $\wedge$ is the wedge product for matrix-valued differential forms [**darling1994**]. Multiplying by ρ^{-1} and using $\psi = \rho^{-1} \, \mathrm{d}\rho$ results in the Maurer-Cartan equation,

$$\mathrm{d}\psi + \psi \wedge \psi = 0. \tag{4}$$

This gives the commutation relation,

$$(\mathrm{d}\psi + \psi \wedge \psi)\,(\partial_s, \partial_t) = \partial_s \psi_t - \partial_t \psi_s + [\psi_s, \psi_t] = 0, \tag{5}$$

where $[\psi_s, \psi_t] := \psi_s \psi_t - \psi_t \psi_s$ is the matrix commutator. Expressing (5) in terms of $\hat{\boldsymbol{\Omega}}, \boldsymbol{U}, \hat{\boldsymbol{\omega}}$, and $\boldsymbol{u}$, and using the fact that $\widehat{\boldsymbol{a} \times \boldsymbol{b}} = \left[\hat{\boldsymbol{a}}, \hat{\boldsymbol{b}}\right]$, for any $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^3$, gives Equations (**??**) and (**??**) found in the main text.

2.1.1 Shape and spatial orientation

Equation (**??**) is equivalent to

$$\rho(0, t + \xi) = \rho(0, t) \mathrm{Pexp} \int_0^{\xi} \gamma(\xi')\, \mathrm{d}\xi'\,, \tag{6}$$

with $\gamma(\xi) := \psi_t(0, t+\xi)$. The ordered exponential in (6) can be expressed as $\mathrm{Pexp} \int_0^{\xi} \gamma(\xi')\, \mathrm{d}\xi' = e^{\theta(\xi)}$, where θ is an $\mathfrak{se}(3)$-valued field related to γ by the Magnus expansion

$$\theta(\xi) = \int_0^{\xi} \gamma(\xi)\, \mathrm{d}\xi - \frac{1}{2} \int_0^{\xi} \int_0^{\xi'} [\gamma(\xi''), \gamma(\xi')]\, \mathrm{d}\xi''\, \mathrm{d}\xi' \,. + \ldots \tag{7}$$

and writing

$$\theta(\xi) = \xi \frac{\gamma(\xi) + \gamma(0)}{2} + O(\xi^3), \tag{8}$$

we have

$$\rho(0, t + \Delta t) = \rho(0, t) \exp\left(\Delta t \frac{\psi_t(0, t + \Delta t) + \psi_t(0, t)}{2} \right) + O(\Delta t^3). \tag{9}$$

By a similar argument, we find

$$\rho(s + \Delta s, t) = \rho(s, t) \exp\left(\Delta s \frac{\psi_s(s + \Delta s, t) + \psi_s(s, t)}{2} \right) + O(\Delta s^3). \tag{10}$$

Matrix exponentials in (9)-(10) are computed using the closed form expression for the exponential map on $SE(3)$,

$$\exp \begin{pmatrix} \hat{\boldsymbol{\alpha}} & \boldsymbol{b} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} e^{\hat{\boldsymbol{\alpha}}} & A\boldsymbol{b} \\ 0 & 1 \end{pmatrix}, \tag{11}$$

where $e^{\hat{\boldsymbol{\alpha}}} = 1 + \dfrac{\sin \alpha}{\alpha} \hat{\boldsymbol{\alpha}} + \dfrac{1 - \cos \alpha}{\alpha^2} \hat{\boldsymbol{\alpha}}^2$, $A = 1 + \dfrac{1 - \cos \alpha}{\alpha^2} \hat{\boldsymbol{\alpha}} + \dfrac{\alpha - \sin \alpha}{\alpha^3} \hat{\boldsymbol{\alpha}}^2$, and $\alpha = |\boldsymbol{\alpha}|$.

2.2 Derivation of Euler-Poincaré equations

The Euler-Poincaré equations, which govern local force and moment balance along the body, are obtained using a constrained variational principal. The variation $\delta\rho := \partial_\xi \tilde{\rho}(\xi)|_{\xi=0}$ is given by the derivative of a field, $\tilde{\rho} = \tilde{\rho}(s,t,\xi)$, which satisfies $\tilde{\rho}(s,t,0) = \rho(s,t)$ [**lrh89**]. The associated 1-form $\tilde{\psi} = \tilde{\psi}_s \,\mathrm{d}s + \tilde{\psi}_t \,\mathrm{d}t + \tilde{\psi}_\xi \,\mathrm{d}\xi := \tilde{\rho}^{-1}\,\mathrm{d}\tilde{\rho}$ now includes a ξ-component which acts as the generator of variations $\delta\rho = \rho\tilde{\psi}_\xi\Big|_{\xi=0} := \rho\psi_\xi$. It follows directly from the definitions of $(\psi, \tilde{\psi}, \tilde{\rho})$ that $\tilde{\psi}_s(s,t,0) = \psi_s(s,t)$, so, the variation $\delta\psi_s(s,t) = \partial_\xi \tilde{\psi}_s(s,t,\xi)\Big|_{\xi=0}$ is constrained by the Maurer-Cartan equation,

$$\left(\mathrm{d}\tilde{\psi} + \tilde{\psi}\wedge\tilde{\psi}\right)(\partial_s, \partial_\xi)\Big|_{\xi=0} = \delta\psi_s - \partial_s\psi_\xi + [\psi_s, \psi_\xi] = 0. \tag{12}$$

In terms of angular and linear components of $\psi_\xi = \begin{pmatrix} \hat{\boldsymbol{\omega}}_\xi & \boldsymbol{u}_\xi \\ 0 & 0 \end{pmatrix}$, equation (12) reads

$$\delta\boldsymbol{\Omega} = \partial_s\boldsymbol{\omega}_\xi + \boldsymbol{\Omega}\times\boldsymbol{\omega}_\xi \tag{13}$$

$$\delta\boldsymbol{U} = \partial_s\boldsymbol{u}_\xi + \boldsymbol{\Omega}\times\boldsymbol{u}_\xi - \boldsymbol{\omega}_\xi\times\boldsymbol{U}. \tag{14}$$

We define the inner product for $\mathfrak{se}(3)$-valued matrices $\theta = \begin{pmatrix} \hat{\boldsymbol{\omega}}_\theta & \boldsymbol{u}_\theta \\ 0 & 0 \end{pmatrix}$ and $\phi = \begin{pmatrix} \hat{\boldsymbol{\omega}}_\phi & \boldsymbol{u}_\phi \\ 0 & 0 \end{pmatrix}$ by $\langle\theta,\phi\rangle := \boldsymbol{\omega}_\theta\cdot\boldsymbol{\omega}_\phi + \boldsymbol{u}_\theta\cdot\boldsymbol{u}_\phi$ and the *coadjoint operator*, $\mathrm{ad}^*_{\psi_s}$, by $\langle \mathrm{ad}^*_{\psi_s}\theta, \phi\rangle := \langle\theta, [\psi_s,\phi]\rangle$. In terms of angular and linear components, we find

$$\mathrm{ad}^*_{\psi_s}\theta = -\begin{pmatrix} \widehat{\boldsymbol{\Omega}\times\boldsymbol{\omega}}_\theta + \widehat{\boldsymbol{U}\times\boldsymbol{u}}_\theta & \boldsymbol{\Omega}\times\boldsymbol{u}_\theta \\ 0 & 0 \end{pmatrix}. \tag{15}$$

With an internal energy density $\mathcal{E} = \mathcal{E}(Q, \boldsymbol{r}, \boldsymbol{\Omega}, \boldsymbol{U})$, the variation of $E = \int_{-L/2}^{L/2} \mathcal{E}\,\mathrm{d}s$ generated by ψ_ξ is

$$\delta E = \int_{-L/2}^{L/2} \left\langle -\partial_s\frac{\delta\mathcal{E}}{\delta\psi_s} + \mathrm{ad}^*_{\psi_s}\frac{\delta\mathcal{E}}{\delta\psi_s} + \rho^{-1}\frac{\delta\mathcal{E}}{\delta\rho}, \psi_\xi\right\rangle \mathrm{d}s + \left\langle \frac{\delta\mathcal{E}}{\delta\psi_s}, \psi_\xi\right\rangle\Bigg|_{-L/2}^{L/2}. \tag{16}$$

Balancing $\delta\mathcal{E}$ with the virtual work, $\mathcal{W} = \int_{-L/2}^{L/2} \langle \mathcal{N}, \psi_\xi \rangle \, \mathrm{d}s$, done by nonconservative stresses, $\mathcal{N} = \begin{pmatrix} \hat{\boldsymbol{M}} & \boldsymbol{F} \\ 0 & 0 \end{pmatrix}$, results in the Euler-Poincaré equations,

$$-\partial_s \frac{\delta\mathcal{E}}{\delta\psi_s} + \mathrm{ad}^*_{\psi_s} \frac{\delta\mathcal{E}}{\delta\psi_s} + \rho^{-1} \frac{\delta\mathcal{E}}{\delta\rho} = \mathcal{N}. \tag{17}$$

Separating (17) into its linear and angular parts gives

$$-\left(\partial_s + \boldsymbol{\Omega}\times\right) \boldsymbol{\nabla}_{\boldsymbol{\Omega}}\mathcal{E} - \boldsymbol{U} \times \boldsymbol{\nabla}_{\boldsymbol{U}}\mathcal{E} + Q^T \left(\boldsymbol{q}_i \times \boldsymbol{\nabla}_{\boldsymbol{q}_i}\mathcal{E}\right) = \boldsymbol{M}, \tag{18}$$

$$-\left(\partial_s + \boldsymbol{\Omega}\times\right) \boldsymbol{\nabla}_{\boldsymbol{U}}\mathcal{E} + Q^T \boldsymbol{\nabla}_{\boldsymbol{r}}\mathcal{E} = \boldsymbol{F}, \tag{19}$$

where the repeated index implies summation. Natural boundary conditions are given by

$$\left(\boldsymbol{\nabla}_{\boldsymbol{\Omega}}\mathcal{E}, \boldsymbol{\nabla}_{\boldsymbol{U}}\mathcal{E}\right)\big|_{s=\pm L/2} = (0, 0). \tag{20}$$

3 Surface geometry

Let a membrane $M = \phi^{-1}(0)$ be defined implicitly as the zero-level set of $\phi : \mathbb{R}^3 \to \mathbb{R}$ with $\phi(\boldsymbol{x}) < 0$ inside and $\phi(\boldsymbol{x}) > 0$ outside of M. Suppose M is endowed with an adapted frame $\rho = (\Xi, \boldsymbol{\Psi})$ such that $\boldsymbol{\xi}_3 = \boldsymbol{\nabla}\phi/|\boldsymbol{\nabla}\phi| = \boldsymbol{n}$ is the surface normal. Define the connection form $\hat{\boldsymbol{\omega}} = \Xi^{-1}\,\mathrm{d}\Xi$, and let $\boldsymbol{\theta} = \Xi^{-1}\,\mathrm{d}\boldsymbol{\Psi}$. In terms of components: of $\boldsymbol{\theta} = \theta^i \boldsymbol{e}_i$, $\hat{\boldsymbol{\omega}} = \omega^i \hat{\boldsymbol{e}}_i$,

$$\boldsymbol{\theta} = \begin{pmatrix} \theta^1 \\ \theta^2 \\ \theta^3 \end{pmatrix} = \begin{pmatrix} \langle \boldsymbol{\xi}_1 | \mathrm{d}\boldsymbol{\Psi} \rangle \\ \langle \boldsymbol{\xi}_2 | \mathrm{d}\boldsymbol{\Psi} \rangle \\ \langle \boldsymbol{\xi}_3 | \mathrm{d}\boldsymbol{\Psi} \rangle = 0 \end{pmatrix}. \tag{21}$$

The first structure equation, $\mathrm{d}\boldsymbol{\theta} = -\hat{\boldsymbol{\omega}} \wedge \boldsymbol{\theta}$, gives

$$\begin{pmatrix} \mathrm{d}\theta^1 \\ \mathrm{d}\theta^2 \\ 0 \end{pmatrix} = \begin{pmatrix} \omega^3 \wedge \theta^2 \\ -\omega^3 \wedge \theta^1 \\ \omega^2 \wedge \theta^1 - \omega^1 \wedge \theta^2 \end{pmatrix}, \tag{22}$$

and the second, $\mathrm{d}\hat{\boldsymbol{\omega}} = -\hat{\boldsymbol{\omega}} \wedge \hat{\boldsymbol{\omega}}$

$$\begin{pmatrix} \mathrm{d}\omega^1 \\ \mathrm{d}\omega^2 \\ \mathrm{d}\omega^3 \end{pmatrix} = \begin{pmatrix} -\omega^2 \wedge \omega^3 \\ -\omega^3 \wedge \omega^1 \\ -\omega^1 \wedge \omega^2 \end{pmatrix} \tag{23}$$

Curvature tensor K_i^j and shape operator $\mathrm{d}\boldsymbol{n}$

$$\begin{aligned}
\boldsymbol{n} &= \boldsymbol{n}(\boldsymbol{\Psi}) && (24)\\
\mathrm{d}\boldsymbol{n} &= \omega^2\boldsymbol{\xi}_1 - \omega^1\boldsymbol{\xi}_2 && (25)\\
&= \mathrm{d}\boldsymbol{\Psi}\cdot\boldsymbol{\nabla}\boldsymbol{n} && (26)\\
&= \left(\theta^i\boldsymbol{\xi}_i\right)\cdot\boldsymbol{\nabla}\boldsymbol{n} && (27)\\
&= -K_i^j\theta^i\boldsymbol{\xi}_j && (28)\\
&= -\left(K_1^1\theta^1 + K_2^1\theta^2\right)\boldsymbol{\xi}_1 - \left(K_1^2\theta^1 + K_2^2\theta^2\right)\boldsymbol{\xi}_2 && (29)
\end{aligned}$$

$$\begin{aligned}
\omega^1 &= K_1^2\theta^1 + K_2^2\theta^2 && (30)\\
\omega^2 &= -K_1^1\theta^1 - K_2^1\theta^2 && (31)
\end{aligned}$$

Gaussian curvature K

$$\begin{aligned}
\omega^1\wedge\omega^2 &= \left(K_1^1K_2^2 - K_1^2K_2^1\right)\theta^1\wedge\theta^2 && (32)\\
&= K\theta^1\wedge\theta^2 && (33)
\end{aligned}$$

Mean curvature H

$$\begin{aligned}
\boldsymbol{\theta}\cdot\boldsymbol{\omega} &= \theta^1\wedge\omega^1 + \theta^2\wedge\omega^2 && (34)\\
&= \left(K_1^1 + K_2^2\right)\theta^1\wedge\theta^2 && (35)\\
&= H\theta^1\wedge\theta^2 && (36)
\end{aligned}$$

4 Surfacebound filament

Consider a framed curve $(\boldsymbol{r}, Q)$ represented by

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \boldsymbol{q}_3 & \boldsymbol{r} \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{37}$$

with $\partial_s\boldsymbol{r} = \boldsymbol{q}_1$, and which is adapted to M

$$\begin{aligned}
& && (38)\\
\phi(\boldsymbol{r}) &= 0 && (39)\\
\boldsymbol{q}_3 &= \boldsymbol{n}(\boldsymbol{r}). && (40)
\end{aligned}$$

4.1 Energy

Let the internal energy of the filament be given by

$$E = \int \mathcal{E}\,\mathrm{d}s + \mathcal{U} \tag{41}$$

$$\mathcal{E} = \frac{1}{2}\left(\boldsymbol{\Omega} - \sigma\right) \cdot B\left(\boldsymbol{\Omega} - \sigma\right) + \boldsymbol{\lambda} \cdot \boldsymbol{U} + \nu\phi/|\boldsymbol{\nabla}\phi| + \mu \boldsymbol{q}_2 \cdot \boldsymbol{n} \tag{42}$$

where $\mathcal{U} = \frac{1}{2}\int\int G(\boldsymbol{r}(s), \boldsymbol{r}(s'))\,\mathrm{d}s'\,\mathrm{d}s$ is a symmetric kernel associated with nonlocal steric interactions. The first variation of the energy is given by

$$\delta E[\rho\chi] = \int \partial_s \left\langle \frac{\delta\mathcal{E}}{\delta\psi}\middle|\chi\right\rangle + \left\langle -\partial_s \frac{\delta\mathcal{E}}{\delta\psi} + \mathrm{ad}^*_\psi \frac{\delta\mathcal{E}}{\delta\psi}\middle|\chi\right\rangle + \left\langle \frac{\delta\mathcal{E}}{\delta\rho}\middle|\rho\chi\right\rangle \mathrm{d}s + \delta\mathcal{U}[\rho\chi] \tag{43}$$

$$= \int \partial_s \left\langle \frac{\delta\mathcal{E}}{\delta\psi}\middle|\chi\right\rangle + \left\langle -\partial_s \frac{\delta\mathcal{E}}{\delta\psi} + \mathrm{ad}^*_\psi \frac{\delta\mathcal{E}}{\delta\psi} + \rho^{-1}\frac{\delta\mathcal{E}}{\delta\rho}\middle|\chi\right\rangle \mathrm{d}s + \delta\mathcal{U}[\rho\chi] \tag{44}$$

where $\left\langle \frac{\delta\mathcal{E}}{\delta\rho}\middle|\rho\chi\right\rangle = \left\langle \rho^{-1}\frac{\delta\mathcal{E}}{\delta\rho}\middle|\chi\right\rangle$ by left invariance of $\langle\cdot|\cdot\rangle$. External work:

$$\mathcal{N} = \begin{pmatrix} -\zeta_r\omega_1\hat{\boldsymbol{e}}_1 & -\zeta\boldsymbol{u} \\ 0 & 0 \end{pmatrix} \tag{45}$$

$$W = \int \langle \mathcal{N}|\chi\rangle\,\mathrm{d}s \tag{46}$$

$$\frac{\delta\mathcal{E}}{\delta\psi} = \begin{pmatrix} \hat{\boldsymbol{m}} & \boldsymbol{\lambda} \\ 0 & 0 \end{pmatrix} \tag{47}$$

Define $\mathcal{C} = \nu\phi/|\boldsymbol{\nabla}\phi| + \mu\boldsymbol{q}_2 \cdot \boldsymbol{n}$, $\delta Q = Q\hat{\boldsymbol{\theta}}$, and $D\boldsymbol{n} = \boldsymbol{\nabla}\boldsymbol{n}^T$. Then using $\phi(\boldsymbol{r}) = 0$, $\delta\boldsymbol{q}_2 = (Q\boldsymbol{\theta}) \times \boldsymbol{q}_2$, and $\boldsymbol{q}_2 \times \boldsymbol{n} = \boldsymbol{q}_1$, we have

$$\delta\mathcal{C} = \nu\boldsymbol{n} \cdot \delta\boldsymbol{r} + \mu\boldsymbol{q}_2 \cdot D\boldsymbol{n} \cdot \delta\boldsymbol{r} + [(Q\boldsymbol{\theta}) \times \boldsymbol{q}_1] \cdot \boldsymbol{n} \tag{48}$$

$$= (\nu\boldsymbol{n} + \mu\boldsymbol{q}_2 \cdot D\boldsymbol{n}) \cdot \delta\boldsymbol{r} + (\mu\boldsymbol{e}_1) \cdot \boldsymbol{\theta}. \tag{49}$$

From left invariance of the inner product on $SO(3)$

$$\left\langle \hat{\boldsymbol{e}}_1 \middle| \hat{\boldsymbol{\theta}}\right\rangle_{T_1 SO(3)} = \left\langle Q\hat{\boldsymbol{e}}_1 \middle| Q\hat{\boldsymbol{\theta}}\right\rangle_{T_Q SO(3)}, \tag{50}$$

it follows that

$$\delta\mathcal{C} = \left\langle \begin{pmatrix} \mu Q\hat{\boldsymbol{e}}_1 & \nu\boldsymbol{n} + \mu\boldsymbol{q}_2 \cdot D\boldsymbol{n} \\ 0 & 0 \end{pmatrix} \middle| \rho\chi \right\rangle \tag{51}$$

$$\frac{\delta\mathcal{C}}{\delta\rho} = \begin{pmatrix} \mu Q\hat{\boldsymbol{e}}_1 & \nu\boldsymbol{n} + \mu\boldsymbol{q}_2 \cdot D\boldsymbol{n} \\ 0 & 0 \end{pmatrix} \tag{52}$$

Self-energy

$$\delta\mathcal{U}[\rho\xi] = \int \left[\int \frac{\partial G}{\partial \boldsymbol{r}(s)} \left(\boldsymbol{r}(s), \boldsymbol{r}(s')\right) \mathrm{d}s' \right] \cdot \delta\boldsymbol{r}(s)\, \mathrm{d}s \tag{53}$$

Steric stress

$$\mathcal{F} = \begin{pmatrix} 0 & \int \frac{\partial G}{\partial \boldsymbol{r}(s)} \left(\boldsymbol{r}(s), \boldsymbol{r}(s')\right) ds' \\ 0 & 0 \end{pmatrix} \tag{54}$$

4.2 stress balance

$$-\partial_s \frac{\delta\mathcal{E}}{\delta\psi} + \mathrm{ad}^*_\psi \frac{\delta\mathcal{E}}{\delta\psi} + \rho^{-1}\frac{\delta\mathcal{E}}{\delta\rho} + \rho^{-1}\mathcal{F} = \mathcal{N} \tag{55}$$

$$\left\langle \frac{\delta\mathcal{E}}{\delta\psi} \middle| \chi \right\rangle \Bigg|_{s=0}^{L} \tag{56}$$

$$\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m} + \boldsymbol{U} \times \boldsymbol{\lambda} - \mu\boldsymbol{e}_1 = \zeta_r \omega_1 \boldsymbol{e}_1 \tag{57}$$

$$\partial_s \boldsymbol{\lambda} + \boldsymbol{\Omega} \times \boldsymbol{\lambda} - \nu\boldsymbol{e}_3 - \mu Q^T \boldsymbol{\nabla}\boldsymbol{n} Q \boldsymbol{e}_2 + \boldsymbol{f} = \zeta\boldsymbol{u} \tag{58}$$

$$\partial_t \boldsymbol{\Omega} = \partial_s \boldsymbol{\omega} + \boldsymbol{\Omega} \times \boldsymbol{\omega} \tag{59}$$

$$\partial_s \boldsymbol{u} + \boldsymbol{\Omega} \times \boldsymbol{u} = \boldsymbol{\omega} \times \boldsymbol{e}_1 \tag{60}$$

$$\partial_s \boldsymbol{r} = Q\boldsymbol{e}_1 \tag{61}$$

$$\phi(\boldsymbol{r}) = 0 \tag{62}$$

$$\boldsymbol{q}_2 \cdot \boldsymbol{n}(\boldsymbol{r}) = 0 \tag{63}$$

4.3 Elimination of Lagrange multipliers

Crossing moment balance with $\boldsymbol{e}_1$ and defining $\lambda := \boldsymbol{e}_1 \cdot \boldsymbol{\lambda}$ gives

$$\boldsymbol{\lambda} = \lambda\boldsymbol{e}_1 + \boldsymbol{e}_1 \times (\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m}) \tag{64}$$

$$\lambda_2 = -\partial_s m_3 - \Omega_1 m_2 + \Omega_2 m_1 \tag{65}$$
$$= -\partial_s m_3 - K_{11} m_1 - K_{12} m_2 \tag{66}$$

$$\lambda_3 = \partial_s m_2 + \Omega_3 m_1 - \Omega_1 m_3 \tag{67}$$
$$= \partial_s m_2 + \Omega_3 m_1 - K_{12} m_3 \tag{68}$$

The remaining $\boldsymbol{e}_1$ component of moment balance gives

$$\mu = \boldsymbol{e}_1 \cdot (\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m}) - \zeta_r \omega_1 \tag{69}$$
$$= \boldsymbol{e}_1 \cdot (\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m}) - \zeta_r K_{12} u_1 - \zeta_r K_{22} u_2 \tag{70}$$

Then a miracle happens and you get

$$\begin{pmatrix} \partial_s \lambda + \boldsymbol{\Omega} \cdot \partial_s \boldsymbol{m} + f_1 \\ \boldsymbol{e}_2 \cdot (\partial_s \boldsymbol{\lambda} + \boldsymbol{\Omega} \times \boldsymbol{\lambda}) + K_{22} \boldsymbol{e}_1 \cdot (\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m}) + f_2 \end{pmatrix} = \begin{pmatrix} \zeta_1 + \zeta_r K_{12}^2 & \zeta_r K_{12} K_{22} \\ \zeta_r K_{12} K_{22} & \zeta_2 + \zeta_r K_{22}^2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \tag{71}$$

where components of the steric force are given by

$$f_1(s) = -\boldsymbol{q}_1(s) \cdot \int \frac{\partial G}{\partial \boldsymbol{r}(s)} \left(\boldsymbol{r}(s), \boldsymbol{r}(s')\right) ds' = -\partial_s \int G\left(\boldsymbol{r}(s), \boldsymbol{r}(s')\right) ds' \tag{72}$$

$$f_2(s) = -\boldsymbol{q}_2(s) \cdot \int \frac{\partial G}{\partial \boldsymbol{r}(s)} \left(\boldsymbol{r}(s), \boldsymbol{r}(s')\right) ds' \tag{73}$$

Define the effective mobility tensor

$$M = \frac{1}{\zeta_1 \zeta_2 + \zeta_1 \zeta_r K_{22}^2 + \zeta_2 \zeta_r K_{12}^2} \begin{pmatrix} \zeta_2 + \zeta_r K_{22}^2 & -\zeta_r K_{12} K_{22} \\ -\zeta_r K_{12} K_{22} & \zeta_1 + \zeta_r K_{12}^2 \end{pmatrix} \tag{74}$$
$$= \sum_{i,j=1}^{2} M_{ij} \boldsymbol{e}_i \boldsymbol{e}_j^T \tag{75}$$

and force density

$$\boldsymbol{F} = \begin{pmatrix} \partial_s \lambda + \boldsymbol{\Omega} \cdot \partial_s \boldsymbol{m} + f_1 \\ \boldsymbol{e}_2 \cdot (\partial_s \boldsymbol{\lambda} + \boldsymbol{\Omega} \times \boldsymbol{\lambda}) + K_{22} \boldsymbol{e}_1 \cdot (\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m}) + f_2 \end{pmatrix} \tag{76}$$
$$= \begin{pmatrix} \partial_s \lambda + \boldsymbol{\Omega} \cdot \partial_s \boldsymbol{m} + f_1 \\ \Omega \lambda + \boldsymbol{e}_2 \cdot (\partial_s \boldsymbol{\lambda}_\perp + \boldsymbol{\Omega} \times \boldsymbol{\lambda}_\perp) + K_{22} \boldsymbol{e}_1 \cdot (\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m}) + f_2 \end{pmatrix} \tag{77}$$
$$= \begin{pmatrix} \partial_s \lambda + \boldsymbol{\Omega} \cdot \partial_s \boldsymbol{m} + f_1 \\ \Omega_3 \lambda + \partial_s \lambda_2 - \Omega_1 \lambda_3 + K_{22} \partial_s m_1 - K_{22} K_{11} m_3 - K_{22} \Omega_3 m_2 + f_2 \end{pmatrix} \tag{78}$$
$$= F_1 \boldsymbol{e}_1 + F_2 \boldsymbol{e}_2 \tag{79}$$

$$F_2 = -\partial_s^2 m_3 + (\lambda - K_{12}m_1 - K_{22}m_2)\,\Omega_3 - \kappa_G m_3 \tag{80}$$
$$- \partial_s\,(K_{11}m_1 + K_{12}m_2) - \Omega_1\partial_s m_2 + K_{22}\partial_s m_1 + f_2 \tag{81}$$

We find

$$\boldsymbol{u} = M\boldsymbol{F} \tag{82}$$

The structure equations and constraints require,

$$\partial_s u_1 - \Omega_3 u_2 = 0 \tag{83}$$
$$\partial_s u_2 + \Omega_3 u_1 = \omega_3 \tag{84}$$

$$\partial_t \Omega_3 = \partial_s \omega_3 + \Omega_1\omega_2 - \Omega_2\omega_1 \tag{85}$$
$$= \partial_s \omega_3 + \kappa_G u_2 \tag{86}$$
$$= \partial_s^2 u_2 + \partial_s\Omega_3 u_1 + \Omega_3\partial_s u_1 + \kappa_G u_2 \tag{87}$$
$$= \partial_s^2 u_2 + \left(\Omega_3^2 + \kappa_G\right) u_2 + \partial_s\Omega_3 u_1 \tag{88}$$
$$\tag{89}$$

where $\kappa_G = K_{11}K_{22} - K_{12}^2$ is the Gaussian curvature at $\boldsymbol{r}$.
Tension equation:

$$\partial_s u_1 - \Omega_3 u_2 = 0 \tag{90}$$

Geodesic curvature equation:

$$\partial_t \Omega_3 = \partial_s^2 u_2 + \left(\Omega_3^2 + \kappa_G\right) u_2 + \partial_s\Omega_3 u_1 \tag{91}$$

4.4 Shape evolution equations

The state is given by $(\Omega, \lambda, \boldsymbol{r}, \boldsymbol{q}) := (\Omega_3, \lambda, \boldsymbol{r}, \boldsymbol{q}_1)$.

$$\Omega_1, \Omega_2 = \Omega_1(\boldsymbol{r}, \boldsymbol{q}), \Omega_2(\boldsymbol{r}, \boldsymbol{q}) \tag{92}$$
$$\lambda_2 = \lambda_2(\partial_s\Omega, \boldsymbol{r}, \boldsymbol{q}) \tag{93}$$
$$\lambda_3 = \lambda_3(\Omega, \boldsymbol{r}, \boldsymbol{q}) \tag{94}$$
$$M = M(\boldsymbol{r}, \boldsymbol{q}) \tag{95}$$
$$F_1 = F_1(\Omega, \partial_s\Omega, \partial_s\lambda, \boldsymbol{r}, \boldsymbol{q}; \boldsymbol{r}) \tag{96}$$
$$F_2 = F_2(\Omega, \partial_s^2\Omega, \lambda, \boldsymbol{r}, \boldsymbol{q}; \boldsymbol{r}) \tag{97}$$
$$\boldsymbol{u} = \boldsymbol{u}(\Omega, \partial_s\Omega, \partial_s^2\Omega, \lambda, \partial_s\lambda, \boldsymbol{q}, \boldsymbol{r}; \boldsymbol{r}) \tag{98}$$
$$\omega = \omega(\Omega, \partial_s\Omega, \partial_s^2\Omega, \partial_s^3\Omega, \lambda, \partial_s\lambda, \partial_s^2\lambda, \boldsymbol{q}, \boldsymbol{r}; \boldsymbol{r}) \tag{99}$$

where "$(\ldots;\boldsymbol{r})$" indicates nonlocal dependence on $\boldsymbol{r}$. The tension equation takes the form

$$\Lambda(\Omega, \partial_s\Omega, \partial_s^2\Omega, \partial_s^3\Omega, \lambda, \partial_s\lambda, \partial_s^2\lambda, \boldsymbol{q}, \boldsymbol{r}; \boldsymbol{r}) = \partial_s u_1 - \Omega u_2 = 0 \tag{100}$$

and the geodesic curvature equation takes the form

$$\partial_t\Omega - \dot{\Omega}(\Omega, \ldots, \partial_s^4\Omega, \lambda, \ldots, \partial_s^3\lambda, \boldsymbol{q}, \boldsymbol{r}; \boldsymbol{r}) = 0 \tag{101}$$

$$\dot{\Omega} := \partial_s\omega + \kappa_G u_2 \tag{102}$$

4.5 steric force

$$\boldsymbol{f}(s) = -\int_0^L \frac{\partial_p g(p(s,\xi))\boldsymbol{p}(s,\xi)}{p(s,\xi)}\,\mathrm{d}\xi \tag{103}$$

where $\boldsymbol{p}(s,\xi) := Q(s)^{-1}\left(\boldsymbol{r}(s) - \boldsymbol{r}(\xi)\right)$

4.6 Boundary conditions

$$\delta\rho = \rho\chi \tag{104}$$

$$\chi = \begin{pmatrix} \hat{\boldsymbol{\theta}} & \boldsymbol{v} \\ 0 & 0 \end{pmatrix}, \tag{105}$$

$$\left\langle \frac{\delta\mathcal{E}}{\delta\psi}\middle|\chi\right\rangle = \boldsymbol{m}\cdot\boldsymbol{\theta} + \boldsymbol{\lambda}\cdot\boldsymbol{v} = 0 \tag{106}$$

4.6.1 Clamped end $s = 0$

Velocities at the clamped end must be zero, $u_1 = u_2 = 0$ and $\omega = \partial_s u_2 + \Omega u_1 = 0$. Vanishing of u_1, u_2 is equivalent to $\boldsymbol{F} = M^{-1}\boldsymbol{u} = 0$.

$$\boldsymbol{F} = \begin{pmatrix} \partial_s\lambda + \boldsymbol{\Omega}\cdot\partial_s\boldsymbol{m} + f_1 \\ \Omega_3\lambda + \partial_s\lambda_2 - \Omega_1\lambda_3 + K_{22}\partial_s m_1 - K_{22}K_{11}m_3 - K_{22}\Omega_3 m_2 + f_2 \end{pmatrix} = 0 \tag{107}$$

The angular velocity equation $\omega = \partial_s u_2 + \Omega u_1 = \partial_s u_2 = 0$, together with $F_1 = F_2 = 0$, implies

$$M_{12}\partial_s F_1 + M_{22}\partial_s F_2 = 0 \tag{108}$$

for bcs, use $F_1 = F_2 = M_{12}\partial_s F_1 + M_{22}\partial_s F_2 = 0$

$$F_1 = \partial_s \lambda + \boldsymbol{\Omega} \cdot \partial_s \boldsymbol{m} + f_1 = 0 \tag{109}$$

$$F_2 = \Omega_3 \lambda + \partial_s \lambda_2 - \Omega_1 \lambda_3 + K_{22}\partial_s m_1 - K_{22}K_{11}m_3 - K_{22}\Omega_3 m_2 + f_2 = 0 \tag{110}$$

$$-\zeta_r K_{12} K_{22} \partial_s F_1 + \left(\zeta_1 + \zeta_r K_{12}^2\right) \partial_s F_2 = 0 \tag{111}$$

4.6.2 free end $s = L$

$$m_3 = 0 \tag{112}$$

$$\partial_s m_3 + 2K_{12} m_2 + (K_{11} - K_{22})\, m_1 = 0 \tag{113}$$

$$\lambda + K_{12} m_1 - K_{11} m_2 = 0 \tag{114}$$

Proof. Structure equations + surface constraints require

$$\boldsymbol{v} = v_1 \boldsymbol{e}_1 + v_2 \boldsymbol{e}_2 \tag{115}$$

$$\boldsymbol{\theta} = \theta_1 \boldsymbol{e}_1 + \theta_2 \boldsymbol{e}_2 + \theta_3 \boldsymbol{e}_3 \tag{116}$$

$$\theta_1 = K_{12} v_1 + K_{22} v_2 \tag{117}$$

$$\theta_2 = -K_{11} v_1 - K_{12} v_2. \tag{118}$$

Using $\lambda_2 = -\partial_s m_3 - K_{12} m_2 - K_{11} m_1$ gives □

4.7 Structure equations

$$\begin{pmatrix} \partial_s u_1 - \Omega_3 u_2 \\ \partial_s u_2 + \Omega_3 u_1 \end{pmatrix} = \begin{pmatrix} 0 \\ \omega_3 \end{pmatrix} \tag{119}$$

$$\partial_t \Omega_3 = \partial_s \omega_3 + \Omega_1 \omega_2 - \Omega_2 \omega_1 = \partial_s \omega_3 + \kappa_G u_2 \tag{120}$$

4.8 Tension

$$\partial_s^2 \lambda + \frac{\partial_s M_{11}}{M_{11}} \partial_s \lambda + \frac{\partial_s \left(M_{12}\Omega\right) - M_{22}\Omega^2}{M_{11}} \lambda = \frac{-\partial_s \left(M_{1j} Y_j\right) + \Omega M_{j2} Y_j}{M_{11}} \tag{121}$$

where $M_{11} = \frac{\zeta_2 + \zeta_r K_{22}^2}{\zeta_1\zeta_2 + \zeta_1\zeta_r K_{22}^2 + \zeta_2\zeta_r K_{12}^2}$, $M_{12} = \frac{-\zeta_r K_{12} K_{22}}{\zeta_1\zeta_2 + \zeta_1\zeta_r K_{22}^2 + \zeta_2\zeta_r K_{12}^2}$, $M_{22} = \frac{\zeta_1 + \zeta_r K_{12}^2}{\zeta_1\zeta_2 + \zeta_1\zeta_r K_{22}^2 + \zeta_2\zeta_r K_{12}^2}$, $Y_1 = \boldsymbol{\Omega} \cdot \partial_s \boldsymbol{m} + f_1$, and $Y_2 = \boldsymbol{e}_2 \cdot (\partial_s \boldsymbol{\lambda}_\perp + \boldsymbol{\Omega} \times \boldsymbol{\lambda}_\perp) + K_{22} \boldsymbol{e}_1 \cdot (\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m}) + f_2$

Proof.

$$\boldsymbol{F} = \begin{pmatrix} \partial_s \lambda + \boldsymbol{\Omega} \cdot \partial_s \boldsymbol{m} + f_1 \\ \Omega\lambda + \boldsymbol{e}_2 \cdot (\partial_s \boldsymbol{\lambda}_\perp + \boldsymbol{\Omega} \times \boldsymbol{\lambda}_\perp) + K_{22} \boldsymbol{e}_1 \cdot (\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m}) + f_2 \end{pmatrix} \tag{122}$$

$$= \begin{pmatrix} \partial_s \lambda \\ \Omega\lambda \end{pmatrix} + \begin{pmatrix} \boldsymbol{\Omega} \cdot \partial_s \boldsymbol{m} + f_1 \\ \boldsymbol{e}_2 \cdot (\partial_s \boldsymbol{\lambda}_\perp + \boldsymbol{\Omega} \times \boldsymbol{\lambda}_\perp) + K_{22} \boldsymbol{e}_1 \cdot (\partial_s \boldsymbol{m} + \boldsymbol{\Omega} \times \boldsymbol{m}) + f_2 \end{pmatrix} \tag{123}$$

$$= \begin{pmatrix} \partial_s \lambda \\ \Omega\lambda \end{pmatrix} + Y \tag{124}$$

$$M = \frac{1}{\zeta_1\zeta_2 + \zeta_1\zeta_r K_{22}^2 + \zeta_2\zeta_r K_{12}^2} \begin{pmatrix} \zeta_2 + \zeta_r K_{22}^2 & -\zeta_r K_{12} K_{22} \\ -\zeta_r K_{12} K_{22} & \zeta_1 + \zeta_r K_{12}^2 \end{pmatrix} \tag{125}$$

$$\boldsymbol{u} = M\boldsymbol{F} \tag{126}$$

$$= M \begin{pmatrix} \partial_s \lambda \\ \Omega\lambda \end{pmatrix} + MY \tag{127}$$

$$= M \begin{pmatrix} \partial_s \lambda \\ \Omega\lambda \end{pmatrix} + X \tag{128}$$

$$u_1 = M_{11}\partial_s\lambda + M_{12}\Omega\lambda + X_1 \tag{129}$$

$$u_2 = M_{12}\partial_s\lambda + M_{22}\Omega\lambda + X_2 \tag{130}$$

$$\partial_s u_1 - \Omega u_2 = M_{11}\partial_s^2\lambda + \partial_s M_{11}\partial_s\lambda + M_{12}\Omega\partial_s\lambda + \partial_s\left(M_{12}\Omega\right)\lambda + \partial_s X_1 - M_{12}\Omega\partial_s\lambda - M_{22}\Omega^2\lambda - \tag{131}$$

$$= M_{11}\partial_s^2\lambda + \partial_s M_{11}\partial_s\lambda + \left(\partial_s\left(M_{12}\Omega\right) - M_{22}\Omega^2\right)\lambda + \partial_s X_1 - \Omega X_2 \tag{132}$$

□

5 Numerics

5.1 Finite differences

BDF2

$$\langle x| \partial_x = \frac{1}{2h}\langle x-2h| - \frac{2}{h}\langle x-h| + \frac{3}{2h}\langle x| + O(h^2) \tag{133}$$

Non-uniform grid $O(h_1) = O(h_2) = O(h)$

$$\langle x| \partial_x = \frac{h_1}{h_2(h_2-h_1)} \langle x-h_2| - \frac{h_2}{h_1(h_2-h_1)} \langle x-h_1| + \frac{h_1+h_2}{h_1 h_2} \langle x| + O(h^2) \tag{134}$$

Second derivative

$$\langle x| \partial_x^2 = \frac{2\,(h_1-h_2)}{h_3\,(h_1+h_3)\,(h_3-h_2)} \langle x-h_3| + \frac{2\,(h_1-h_3)}{h_2\,(h_1+h_2)\,(h_2-h_3)} \langle x-h_2| \tag{135}$$

$$+ \frac{2\,(h_1-h_2-h_3)}{h_1 h_2 h_3} \langle x| + \frac{2\,(h_2+h_3)}{h_1\,(h_1+h_2)\,(h_1+h_3)} \langle x+h_1| + O(h^2) \tag{136}$$

5.2 *q-r*

$$\psi(\Omega, \boldsymbol{q}, \boldsymbol{r}) = \begin{pmatrix} K_{12}(\boldsymbol{q},\boldsymbol{r})\hat{\boldsymbol{e}}_1 - K_{11}(\boldsymbol{q},\boldsymbol{r})\hat{\boldsymbol{e}}_2 + \Omega\hat{\boldsymbol{e}}_3 & \boldsymbol{e}_1 \\ 0 & 0 \end{pmatrix} \tag{137}$$

$$\Psi := \frac{h}{2}\left(\psi + \tilde{\psi}\right) = \begin{pmatrix} \hat{\boldsymbol{\theta}} & h\boldsymbol{e}_1 \\ 0 & 0 \end{pmatrix} \tag{138}$$

$$\rho = \tilde{\rho} e^{\Psi} + O(h^3) \tag{139}$$

$$\tilde{\rho} e^{\Psi} = \begin{pmatrix} \tilde{Q} & \tilde{\boldsymbol{r}} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e^{\hat{\boldsymbol{\theta}}} & hV\boldsymbol{e}_1 \\ 0 & 1 \end{pmatrix} \tag{140}$$

$$= \begin{pmatrix} \tilde{Q} e^{\hat{\boldsymbol{\theta}}} & \tilde{\boldsymbol{r}} + h\tilde{Q}V\boldsymbol{e}_1 \\ 0 & 1 \end{pmatrix} \tag{141}$$

$A = \sin(\theta)/\theta$, $B = \frac{1-\cos(\theta)}{\theta^2}$, and $C = (1-A)/\theta^2$

$$\boldsymbol{q} = \tilde{Q} e^{\hat{\boldsymbol{\theta}}} \boldsymbol{e}_1 \tag{142}$$

$$= \tilde{Q}\left(\boldsymbol{e}_1 + A\boldsymbol{\theta} \times \boldsymbol{e}_1 + B\boldsymbol{\theta} \times (\boldsymbol{\theta} \times \boldsymbol{e}_1)\right) \tag{143}$$

$$= \begin{pmatrix} \tilde{\boldsymbol{q}} & \tilde{\boldsymbol{n}} \times \tilde{\boldsymbol{q}} & \tilde{\boldsymbol{n}} \end{pmatrix} \begin{pmatrix} 1 - B(\theta_2^2 + \theta_3^2) \\ A\theta_3 + B\theta_1\theta_2 \\ -A\theta_2 + B\theta_1\theta_3 \end{pmatrix} \tag{144}$$

$$\boldsymbol{r} = \tilde{\boldsymbol{r}} + h\tilde{Q}V\boldsymbol{e}_1 \tag{145}$$

$$= \tilde{\boldsymbol{r}} + h\tilde{Q}\left(\boldsymbol{e}_1 + B\boldsymbol{\theta} \times \boldsymbol{e}_1 + C\boldsymbol{\theta} \times (\boldsymbol{\theta} \times \boldsymbol{e}_1)\right) \tag{146}$$

$$= \tilde{\boldsymbol{r}} + h\begin{pmatrix}\tilde{\boldsymbol{q}} & \tilde{\boldsymbol{n}} \times \tilde{\boldsymbol{q}} & \tilde{\boldsymbol{n}}\end{pmatrix}\begin{pmatrix}1 - C(\theta_2^2 + \theta_3^2) \\ B\theta_3 + C\theta_1\theta_2 \\ -B\theta_2 + C\theta_1\theta_3\end{pmatrix} \tag{147}$$

6 Matrix groups

$$\exp : \mathfrak{g} \to G \tag{148}$$

$$\mathrm{d}\exp : T\mathfrak{g} \to TG \tag{149}$$

$$\mathrm{d}\exp_X : T_X\mathfrak{g} \to T_{\exp(Y)}G \tag{150}$$

$$\mathrm{d}\exp_X(Y) = \exp(X)\frac{1 - \exp(-\mathrm{ad}_X)}{\mathrm{ad}_X}Y \tag{151}$$

So, if $g(t) = g_0 \exp(\Theta(t))$ and $\mathrm{d}g = g\Omega$

$$g_0 \exp(\Theta)\Omega = g_0 \,\mathrm{d}\exp_\Theta \mathrm{d}\Theta \tag{152}$$

$$= g_0 \exp(\Theta)\frac{1 - \exp(-\mathrm{ad}_\Theta)}{\mathrm{ad}_\Theta}\mathrm{d}\Theta \tag{153}$$

$$\Rightarrow \Omega = \frac{1 - \exp(-\mathrm{ad}_\Theta)}{\mathrm{ad}_\Theta}\mathrm{d}\Theta \tag{154}$$

$$\mathrm{d}\Theta = \frac{\mathrm{ad}_\Theta}{1 - \exp(-\mathrm{ad}_\Theta)}\Omega \tag{155}$$

$$\mathrm{d}\Theta = \sum_{n=0}^{\infty}\frac{B_n}{n!}\mathrm{ad}_\Theta^n\Omega \tag{156}$$

Where B_n are the Bernoulli numbers with $B = +1/2$:

$$B_m = \sum_{k=0}^{m}\sum_{\ell=0}^{k}(-1)^\ell\binom{k}{\ell}\frac{(\ell+1)^m}{k+1} \tag{157}$$

$$\log(M) = \sum_{j=1}^{\infty} \frac{(-1)^{j+1}}{j} (M-1)^j \tag{158}$$

6.1 BCH

$$\log\left(e^X e^Y\right) = X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}\left([X,[X,Y]] + [Y,[Y,X]]\right) + \dots \tag{159}$$

7 The Special Orthogonal Group SO(3)

Given a rotation matrix $R \in \mathrm{SO}(3)$, the axis of rotation can be obtained by applying the inverse hat map to the antisymmetric matrix $R - R^T$ and the angle of rotation satisfies $\mathrm{tr}(R) = 2\cos(\theta) + 1$.

$$\log(R) = \frac{\theta}{2\sin\theta}\left(R - R^T\right) \tag{160}$$

$$\theta = \arccos\left(\frac{\mathrm{tr}(R) - 1}{2}\right) \tag{161}$$

$$\frac{\theta}{2\sin\theta} = 1/2 + \theta^2/12 + 7\theta^4/720 + 31\theta^6/30240 + O(\theta^8) \tag{162}$$

8 The Special Euclidean Group SE(3)

The *Special Euclidean group* SE(3) is the semi-direct product $SO(3) \ltimes \mathbb{R}^3$ of the three-dimensional *Special Orthogonal group* $SO(3)$ with the three-dimensional *translation group* $\mathbb{R}^3$. The composition law is given by $(R, \boldsymbol{x})(R', \boldsymbol{x}') = (RR', R\boldsymbol{x}' + \boldsymbol{x})$, and inverses by $(R, \boldsymbol{x})^{-1} = (R^{-1}, -R^{-1}\boldsymbol{x})$. A faithful matrix representation is given by

$$\begin{pmatrix} R & \boldsymbol{x} \\ 0 & 1 \end{pmatrix} \tag{163}$$

8.1 exp and log

$$\exp\begin{pmatrix}\hat{\boldsymbol{\theta}} & \boldsymbol{u}\\ 0 & 0\end{pmatrix} = \begin{pmatrix}e^{\hat{\boldsymbol{\theta}}} & V\boldsymbol{u}\\ 0 & 1\end{pmatrix}, \tag{164}$$

where $\theta = |\boldsymbol{\theta}|$, and

$$e^{\hat{\boldsymbol{\theta}}} = 1 + \frac{\sin\theta}{\theta}\hat{\boldsymbol{\theta}} + \frac{1-\cos\theta}{\theta^2}\hat{\boldsymbol{\theta}}^2 \tag{165}$$

$$V = 1 + \frac{1-\cos\theta}{\theta^2}\hat{\boldsymbol{\theta}} + \frac{\theta-\sin\theta}{\theta^3}\hat{\boldsymbol{\theta}}^2 \tag{166}$$

Define $A = \sin(\theta)/\theta$, $B = \frac{1-\cos(\theta)}{\theta^2}$, and $C = (1-A)/\theta^2$, so that

$$e^{\hat{\boldsymbol{\theta}}} = 1 + A\hat{\boldsymbol{\theta}} + B\hat{\boldsymbol{\theta}}^2 \tag{167}$$

$$V = 1 + B\hat{\boldsymbol{\theta}} + C\hat{\boldsymbol{\theta}}^2 \tag{168}$$

$$A = \sum_{j=0}^{\infty}\frac{(-1)^j}{(2j+1)!}\theta^{2j} \tag{169}$$

$$= 1 - \theta^2/3! + \theta^4/5! - \theta^6/7! + O(\theta^8) \tag{170}$$

$$= 1 - \theta^2/6 + \theta^4/120 - \theta^6/540 + O(\theta^8) \tag{171}$$

$$B = \sum_{j=0}^{\infty}\frac{(-1)^j}{(2j+2)!}\theta^{2j} \tag{172}$$

$$= 1/2! - \theta^2/4! + \theta^4/6! - \theta^6/8! + O(\theta^8) \tag{173}$$

$$= 1/2 - \theta^2/24 + \theta^4/720 - \theta^6/40320 + O(\theta^8) \tag{174}$$

$$C = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+3)!}\theta^{2j} \tag{175}$$
$$= 1/3! - \theta^2/5! + \theta^4/7! - \theta^6/9! + O(\theta^8) \tag{176}$$
$$= 1/6 - \theta^2/120 + \theta^4/5040 - \theta^6/362880 + O(\theta^8) \tag{177}$$

$$\log \begin{pmatrix} R & \boldsymbol{x} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{\theta}{2\sin\theta}(R - R^T) & V^{-1}\boldsymbol{x} \\ 0 & 0 \end{pmatrix} \tag{178}$$

$$V^{-1} = 1 - \frac{1}{2}\hat{\boldsymbol{\theta}} + \frac{1}{\theta^2}\left(1 - \frac{A}{2B}\right)\hat{\boldsymbol{\theta}}^2 \tag{179}$$

$$\frac{1}{\theta^2}\left(1 - \frac{A}{2B}\right) = 1/12 + \theta^2/720 + \theta^4/30240 + \theta^6/1209600 + O(\theta^8) \tag{180}$$

8.1.1 2nd Order Magnus integration (spatial)

Quantities evaluated at $s - h$ are written with $\tilde{\rho} := \rho(s-h)$, $\tilde{\boldsymbol{\Omega}} := \hat{\boldsymbol{\Omega}}(s-h)$, etc...

$$\Psi = \begin{pmatrix} \frac{h}{2}\left(\hat{\boldsymbol{\Omega}} + \tilde{\boldsymbol{\Omega}}\right) & h\boldsymbol{e}_1 \\ 0 & 0 \end{pmatrix} \tag{181}$$

To order second order in h,

$$e^{\Psi} = \tilde{\rho}^{-1}\rho \tag{182}$$
$$= \begin{pmatrix} \tilde{Q}^{-1}Q & \tilde{Q}^{-1}(\boldsymbol{r} - \tilde{\boldsymbol{r}}) \\ 0 & 1 \end{pmatrix} \tag{183}$$

$$\Psi = \log \begin{pmatrix} \tilde{Q}^{-1}Q & \tilde{Q}^{-1}(\boldsymbol{r} - \tilde{\boldsymbol{r}}) \\ 0 & 1 \end{pmatrix} \tag{184}$$
$$= \begin{pmatrix} \log\left(\tilde{Q}^{-1}Q\right) & V^{-1}\tilde{Q}^{-1}(\boldsymbol{r} - \tilde{\boldsymbol{r}}) \\ 0 & 0 \end{pmatrix} \tag{185}$$

$$\log\left(\tilde{Q}^{-1}Q\right) - \frac{h}{2}\left(\hat{\boldsymbol{\Omega}} + \tilde{\boldsymbol{\Omega}}\right) = 0 \tag{186}$$

$$V^{-1}\tilde{Q}^{-1}\left(\boldsymbol{r} - \tilde{\boldsymbol{r}}\right) - h\boldsymbol{e}_1 = 0 \tag{187}$$

Define $\hat{\boldsymbol{\theta}} = \log\left(\tilde{Q}^{-1}Q\right) = \frac{\theta}{2\sin\theta}\left(\tilde{Q}^T Q - Q^T\tilde{Q}\right)$, where

$$\theta = \arccos\left(\frac{\mathrm{tr}(\tilde{Q}^T Q) - 1}{2}\right) \tag{188}$$

$$= \arccos\left(1 + \frac{\mathrm{tr}(\tilde{Q}^T Q) - 3}{2}\right) \tag{189}$$

$$\tag{190}$$

$$\boldsymbol{e}_i^T\tilde{Q}^{-1}Q\boldsymbol{e}_j = \tilde{\boldsymbol{q}}_i \cdot \boldsymbol{q}_j \tag{191}$$

$$\mathrm{tr}(\tilde{Q}^{-1}Q) = \tilde{\boldsymbol{q}}_j \cdot \boldsymbol{q}_j \tag{192}$$

$$= \tilde{\boldsymbol{q}} \cdot \boldsymbol{q} + (\tilde{\boldsymbol{n}} \times \tilde{\boldsymbol{q}}) \cdot (\boldsymbol{n} \times \boldsymbol{q}) + \tilde{\boldsymbol{n}} \cdot \boldsymbol{n} \tag{193}$$

Components of $\hat{\boldsymbol{\theta}} = \frac{\theta}{2\sin\theta}\left(\tilde{Q}^T Q - Q^T\tilde{Q}\right)$,

$$\theta_1 = -\frac{\theta}{2\sin\theta}\left(\tilde{\boldsymbol{q}}_2 \cdot \boldsymbol{q}_3 - \tilde{\boldsymbol{q}}_3 \cdot \boldsymbol{q}_2\right) \tag{194}$$

$$\theta_2 = \frac{\theta}{2\sin\theta}\left(\tilde{\boldsymbol{q}}_1 \cdot \boldsymbol{q}_3 - \tilde{\boldsymbol{q}}_3 \cdot \boldsymbol{q}_1\right) \tag{195}$$

$$\theta_3 = -\frac{\theta}{2\sin\theta}\left(\tilde{\boldsymbol{q}}_1 \cdot \boldsymbol{q}_2 - \tilde{\boldsymbol{q}}_2 \cdot \boldsymbol{q}_1\right) \tag{196}$$

$$\theta_i = \frac{\theta}{2\sin\theta}\epsilon_{ijk}\boldsymbol{q}_j \cdot \tilde{\boldsymbol{q}}_k \tag{197}$$

$$= \left(1/2 + \theta^2/12\right)\epsilon_{ijk}\boldsymbol{q}_j \cdot \tilde{\boldsymbol{q}}_k + O(\theta^4) \tag{198}$$

$$\frac{\theta}{2\sin\theta} = 1/2 + \theta^2/12 + 7\theta^4/720 + 31\theta^6/30240 + O(\theta^8) \tag{199}$$

$$\delta\,\mathrm{tr}(\tilde{Q}^{-1}Q) = (\tilde{\boldsymbol{q}}_1 + \tilde{\boldsymbol{q}}_2 \times \boldsymbol{n}) \cdot \delta\boldsymbol{q} + (\tilde{\boldsymbol{q}}_3 - \tilde{\boldsymbol{q}}_2 \times \boldsymbol{q}) \cdot D\boldsymbol{n} \cdot \delta\boldsymbol{r} \tag{200}$$

$$\delta\theta = -\frac{1}{2\sin\theta}\delta\,\mathrm{tr}(\tilde{Q}^{-1}Q) \tag{201}$$

$$\begin{aligned}
\delta(\theta^2) &= 2\theta\delta\theta && (202)\\
&= -\frac{\theta}{\sin\theta}\delta\,\mathrm{tr}(\tilde{Q}^{-1}Q) && (203)\\
&= -\frac{\theta}{\sin\theta}\,(\tilde{\boldsymbol{q}}_1 + \tilde{\boldsymbol{q}}_2 \times \boldsymbol{n}) \cdot \delta\boldsymbol{q} - \frac{\theta}{\sin\theta}\,(\tilde{\boldsymbol{q}}_3 - \tilde{\boldsymbol{q}}_2 \times \boldsymbol{q}) \cdot D\boldsymbol{n} \cdot \delta\boldsymbol{r} && (204)
\end{aligned}$$

8.2 Affine group

8.3 roots

$$\rho : \mathbb{R}^{3+3} \to SE(3) \tag{205}$$

$$\rho(\boldsymbol{q}, \boldsymbol{r}) \begin{pmatrix} & \\ 0 & 1 \end{pmatrix} \tag{206}$$

$$F : \mathbb{R}^{3+3} \to \mathfrak{se}(3) \tag{207}$$

$$F(\boldsymbol{q}, \boldsymbol{r}) = \log\left(e^{\log\left(\tilde{\rho}^{-1}\rho(\boldsymbol{q},\boldsymbol{r})\right)} e^{\frac{h}{2}(\hat{\boldsymbol{\Omega}} + \tilde{\boldsymbol{\Omega}})}\right) \tag{208}$$

χ

8.4 Adjoint action

$$\psi = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{U} \\ 0 & 0 \end{pmatrix}, \chi = \begin{pmatrix} \hat{\boldsymbol{\theta}} & \boldsymbol{v} \\ 0 & 0 \end{pmatrix} \tag{209}$$

$$\langle \psi | \chi \rangle := \boldsymbol{\Omega} \cdot \boldsymbol{\theta} + \boldsymbol{U} \cdot \boldsymbol{v} \tag{210}$$

$$\mathrm{ad}_\psi \chi = \begin{pmatrix} \widehat{\boldsymbol{\Omega} \times \boldsymbol{\theta}} & \boldsymbol{\Omega} \times \boldsymbol{v} - \boldsymbol{\theta} \times \boldsymbol{U} \\ 0 & 0 \end{pmatrix} \tag{211}$$

$$\mathrm{ad}^*_\psi \chi = -\begin{pmatrix} \widehat{\boldsymbol{\Omega} \times \boldsymbol{\theta}} + \widehat{\boldsymbol{U} \times \boldsymbol{v}} & \boldsymbol{\Omega} \times \boldsymbol{v} \\ 0 & 0 \end{pmatrix} \tag{212}$$

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix}, \tag{213}$$

$$\mathrm{Ad}_\rho \chi = \rho \chi \rho^{-1} = \begin{pmatrix} \widehat{Q\boldsymbol{\theta}} & Q\boldsymbol{v} - (Q\boldsymbol{\theta}) \times \boldsymbol{r} \\ 0 & 0 \end{pmatrix} \tag{214}$$

$$\mathrm{Ad}^*_\rho \chi = \begin{pmatrix} \widehat{Q^{-1}\boldsymbol{\theta}} - \widehat{Q^{-1}(\boldsymbol{r} \times \boldsymbol{v})} & Q^{-1}\boldsymbol{v} \\ 0 & 0 \end{pmatrix} \tag{215}$$

8.5 Geodesics

geodesics=zero linear/angular acceleration

$$L[\rho, \psi] = \int \langle \psi | \psi \rangle \, \mathrm{d}\xi \tag{216}$$

$$\ldots \tag{217}$$

$$\partial_\xi \boldsymbol{\omega} = 0 \tag{218}$$

$$\partial_\xi \boldsymbol{u} + \boldsymbol{\omega} \times \boldsymbol{u} = 0 \tag{219}$$

9 Helices

Arclength parameterized framed helix with axis $\underline{\boldsymbol{\omega}} = \omega \boldsymbol{e}_z$ through the origin, radius $a = \kappa/\omega^2$, pitch $p = \tau/\omega^2$

$$\boldsymbol{r} = \frac{\kappa}{\omega^2} e^{\omega s \hat{\boldsymbol{e}}_z} \boldsymbol{e}_x + \frac{\tau}{\omega} s \boldsymbol{e}_z \tag{220}$$

$$Q = e^{\omega s \hat{\boldsymbol{e}}_z} Q_0 \tag{221}$$

$$\boldsymbol{q}_z = \partial_s \boldsymbol{r} = \frac{\kappa}{\omega} e^{\omega s \hat{\boldsymbol{e}}_z} \boldsymbol{e}_y + \frac{\tau}{\omega} \boldsymbol{e}_z \tag{222}$$

$\boldsymbol{q}_z(0) = sin(\alpha)\boldsymbol{e}_y + \cos(\alpha)\boldsymbol{e}_z$

10 Useful things

$$K(Q, \boldsymbol{r}) = \begin{pmatrix} K_{11} & K_{12} \\ K_{12} & K_{22} \end{pmatrix} = \begin{pmatrix} -\boldsymbol{q}_1 \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{q}_1 & -\boldsymbol{q}_1 \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{q}_2 \\ -\boldsymbol{q}_1 \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{q}_2 & -\boldsymbol{q}_2 \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{q}_2 \end{pmatrix} = \begin{pmatrix} -\Omega_2 & \Omega_1 \\ \Omega_1 & -\boldsymbol{q}_2 \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{q}_2 \end{pmatrix} \tag{223}$$

$$\Omega_1 = -\boldsymbol{q}_1 \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{q}_2 = K_{12} \tag{224}$$

$$\Omega_2 = \boldsymbol{q}_1 \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{q}_1 = -K_{11} \tag{225}$$

$$\omega_1 = -\partial_t \boldsymbol{r} \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{q}_2 = K_{12} u_1 + K_{22} u_2 \tag{226}$$

$$\omega_2 = \partial_t \boldsymbol{r} \cdot \boldsymbol{\nabla n} \cdot \boldsymbol{q}_1 = -K_{11} u_1 - K_{12} u_2 \tag{227}$$

$$\Omega_1 \omega_2 - \Omega_2 \omega_1 = \boldsymbol{n} \cdot [(\partial_s \boldsymbol{r} \cdot \boldsymbol{\nabla n}) \times (\partial_t \boldsymbol{r} \cdot \boldsymbol{\nabla n})] = \kappa_G u_2 \tag{228}$$

$$(\partial_s \boldsymbol{r} \cdot \boldsymbol{\nabla n}) \times (\partial_t \boldsymbol{r} \cdot \boldsymbol{\nabla n}) = \kappa_G u_2 \boldsymbol{n} \tag{229}$$

$$K \hat{\boldsymbol{n}} K = \boldsymbol{\nabla n}^T \hat{\boldsymbol{n}} \boldsymbol{\nabla n} = \kappa_G \hat{\boldsymbol{n}} \tag{230}$$

Which means the Penrose pseudo-inverse of the shape operator is given by

$$K^+ = -\frac{1}{\kappa_G} \hat{\boldsymbol{n}} K \hat{\boldsymbol{n}} \tag{231}$$

$$= \frac{1}{\operatorname{tr}(\hat{\boldsymbol{n}} K \hat{\boldsymbol{n}} K)} \hat{\boldsymbol{n}} K \hat{\boldsymbol{n}} \tag{232}$$

$$(\boldsymbol{q}_1 \cdot \boldsymbol{\nabla n}) \times (\boldsymbol{q}_2 \cdot \boldsymbol{\nabla n}) = \kappa_G \boldsymbol{n} \tag{233}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{n} = -\operatorname{tr}(K) = -2\overline{\kappa} \tag{234}$$

11 The screended Poisson equation and Yukawa potential

$$\left[\nabla^2 - 1/a^2\right]\phi(\mathbf{x}) = -\rho(\mathbf{x}) \tag{235}$$

The potential due to a point charge is called the Yukawa potential

$$\left[\nabla^2 - 1/a^2\right]G(\mathbf{x}) = -\delta(\mathbf{x}) \tag{236}$$

$$G(\mathbf{x}) = \frac{1}{4\pi|\mathbf{x}|}e^{-|\mathbf{x}|/a} \tag{237}$$

$$\begin{aligned} G(\mathbf{x}) &= \frac{1}{4\pi|\mathbf{x}|}e^{-|\mathbf{x}|/a} && (238)\\ &= \frac{1}{(2\pi)^3}\int \frac{1}{|\mathbf{k}|^2 + 1/a^2}e^{i\mathbf{k}\cdot\mathbf{x}}\,\mathrm{d}V(\mathbf{k}) && (239)\\ &= \frac{1}{2\pi^2 r}\int_0^\infty \frac{k\sin(kr)}{k^2 + 1/a^2}\,\mathrm{d}k && (240) \end{aligned}$$

and is the free space Green's function for 235. The solution to 235 with $\phi(\infty) = 0$ is

$$\phi(\mathbf{x}) = \int G(\mathbf{x} - \mathbf{x}')\rho(\mathbf{x}')\,\mathrm{d}V(\mathbf{x}') \tag{241}$$

Electric Field:

$$-\boldsymbol{\nabla}G(\mathbf{x}) = \frac{(r+a)e^{-r/a}}{4\pi a r^3}\mathbf{x} \tag{242}$$

$$\mathbf{E}(\mathbf{x}) = -\boldsymbol{\nabla}\phi \tag{243}$$

The self energy of a distribution ρ with potential ϕ is

$$\begin{aligned} E[\rho] &= \frac{1}{2}\int \rho(\mathbf{x})\phi(\mathbf{x})\,\mathrm{d}V(\mathbf{x}) && (244)\\ &= \frac{1}{2}\int\int \rho(\mathbf{x})\rho(\mathbf{x}')G(\mathbf{x} - \mathbf{x}')\,\mathrm{d}V(\mathbf{x})\,\mathrm{d}V(\mathbf{x}') && (245) \end{aligned}$$

12 Numerics

- geogesic curvature – Ω

- tension – λ

13 Steric interactions

Place charges at points $s_0 = 0, \ell, \ldots, n\ell, \ldots, L$ along the centerline and let $\boldsymbol{r}_a := \boldsymbol{r}(s_a)$, $\boldsymbol{r}_b := \boldsymbol{r}(s_b)$, $r_{ab} := |\boldsymbol{r}_a - \boldsymbol{r}_b|$, etc... with interaction energy $G_{ab} := G(\boldsymbol{r}_a, \boldsymbol{r}_b)$.

13.1 Lennard-Jones potential

The Lennard-Jones is given by

$$G_{ab} = 4\varepsilon \left[\left(\frac{A}{r_{ab}} \right)^{12} - \left(\frac{A}{r_{ab}} \right)^{6} \right] \tag{246}$$

where ε is the dispersion energy (well depth) and R is the zero energy distance. The minimum energy, $G_{min} = -\varepsilon$, is achieved at $R = 2^{1/6}A$ which we choose so that nearest neighbor interactions have minimum energy when $\boldsymbol{\Omega} = \boldsymbol{\sigma}$. If we define $\sigma = |\boldsymbol{\sigma}|$ and $\cos(\alpha) := \sigma_1/\sigma$, we find

$$R^2 = \ell^2 - \left[\ell^2 - \frac{4}{\sigma^2} \sin^2(\sigma\ell/2) \right] \sin^2(\alpha). \tag{247}$$

$$\boldsymbol{r}(\ell) - \boldsymbol{r}(0) = \int_0^\ell \exp(s\hat{\boldsymbol{\sigma}})\boldsymbol{e}_x \tag{248}$$

$$R^2 = \int_0^\ell \int_0^\ell \boldsymbol{e}_x \cdot \exp((s-s')\hat{\boldsymbol{\sigma}}) \cdot \boldsymbol{e}_x \,\mathrm{d}s\,\mathrm{d}s' \tag{249}$$

$$= \int_0^\ell \int_0^\ell \boldsymbol{e}_x \cdot \left[\boldsymbol{e}_x \cos(|\boldsymbol{\sigma}|(s-s')) + \frac{\boldsymbol{\sigma} \times \boldsymbol{e}_x}{|\boldsymbol{\sigma}|} \sin(|\boldsymbol{\sigma}|(s-s')) + \boldsymbol{\sigma} \frac{\boldsymbol{\sigma} \cdot \boldsymbol{e}_x}{|\boldsymbol{\sigma}|^2} (1 - \cos(|\boldsymbol{\sigma}|(s-s') \right. \tag{250}$$

$$= \int_0^\ell \int_0^\ell \cos(|\boldsymbol{\sigma}|(s-s')) + \frac{(\boldsymbol{\sigma} \cdot \boldsymbol{e}_x)^2}{|\boldsymbol{\sigma}|^2} (1 - \cos(|\boldsymbol{\sigma}|(s-s'))) \,\mathrm{d}s\,\mathrm{d}s' \tag{251}$$

$$= \int_0^\ell \int_0^\ell \frac{(\boldsymbol{\sigma} \cdot \boldsymbol{e}_x)^2}{|\boldsymbol{\sigma}|^2} + \left[1 - \frac{(\boldsymbol{\sigma} \cdot \boldsymbol{e}_x)^2}{|\boldsymbol{\sigma}|^2} \right] \cos(|\boldsymbol{\sigma}|(s-s')) \,\mathrm{d}s\,\mathrm{d}s' \tag{252}$$

$$= \frac{(\boldsymbol{\sigma} \cdot \boldsymbol{e}_x \ell)^2}{|\boldsymbol{\sigma}|^2} + \left[1 - \frac{(\boldsymbol{\sigma} \cdot \boldsymbol{e}_x)^2}{|\boldsymbol{\sigma}|^2} \right] \frac{2 - 2\cos(|\boldsymbol{\sigma}|\ell)}{|\boldsymbol{\sigma}|^2} \tag{253}$$

$$= \frac{\sigma_x^2 \ell^2}{\sigma^2} + \left[1 - \frac{\sigma_x^2}{\sigma^2} \right] \frac{4\sin^2(\sigma\ell/2)}{\sigma^2} \tag{254}$$

13.1.1 continuous distribution

Steric forces $\boldsymbol{F}(s)$ are computed using

$$g(r) := 4\varepsilon \left[\left(\frac{A}{r} \right)^{12} - \left(\frac{A}{r} \right)^{6} \right] \tag{255}$$

$$G(\boldsymbol{x}, \boldsymbol{y}) = g(|\boldsymbol{x} - \boldsymbol{y}|) \tag{256}$$

$$g'(r) = \frac{-24\varepsilon}{A} \left[2 \left(\frac{A}{r} \right)^{13} - \left(\frac{A}{r} \right)^{7} \right] \tag{257}$$

$$\frac{\partial G}{\partial \boldsymbol{x}} (\boldsymbol{x}, \boldsymbol{y}) = g'(|\boldsymbol{x} - \boldsymbol{y}|) \frac{\boldsymbol{x} - \boldsymbol{y}}{|\boldsymbol{x} - \boldsymbol{y}|} \tag{258}$$

$$Q(s)\boldsymbol{f}(s) = -\int_0^L g'(|\boldsymbol{r}(s) - \boldsymbol{r}(\xi)|) \frac{\boldsymbol{r}(s) - \boldsymbol{r}(\xi)}{|\boldsymbol{r}(s) - \boldsymbol{r}(\xi)|} \,\mathrm{d}\xi \tag{259}$$

$$\boldsymbol{f}(s) = -\int_0^L \frac{\partial_p g(p(s,\xi)) \boldsymbol{p}(s,\xi)}{p(s,\xi)} \,\mathrm{d}\xi \tag{260}$$

where $\boldsymbol{p}(s,\xi) := Q(s)^{-1} (\boldsymbol{r}(s) - \boldsymbol{r}(\xi))$